

ECM3412

UNIVERSITY OF EXETER

**SCHOOL OF ENGINEERING, COMPUTER SCIENCE
AND MATHEMATICS**

Nature-Inspired Computation

TWO HOURS

Answer question 1, and two out of the four other questions.

Question 1 is worth 40 marks. Other questions are worth 30 marks each.

**Candidates are advised to spend FORTY-FIVE minutes on question 1
and THIRTY-FIVE minutes on each of the other questions.**

No electronic calculators of any sort are to be used during the course of this examination.

ECM3412 (2010)

Compulsory question

1. (a) Write pseudocode for an *evolutionary algorithm* (EA). For each stage, explain how the operation affects the ability of the EA to find a near-optimal solution.

(8 marks)

- (b) Write pseudocode for the *hillclimbing algorithm* and explain why it is not desirable to use this algorithm on most large real-world problems.

(6 marks)

- (c) Cellular Automata have three main properties that distinguish them from other algorithms. Name these and briefly explain what is meant by each.

(6 marks)

- (d) Working for a chemical company, you conduct an experiment to find a compound with small size but high effectiveness. You find that compound A had higher effectiveness than B and C but was larger than compound C. Compound B had higher effectiveness and was smaller than C. From what you know of the domination criterion in multi-objective optimisation, state which compound(s) are

- (i) Dominated
- (ii) Non-Dominated

(4 marks)

Which of these compounds would you recommend to the company? Why?

(2 marks)

- (e) Briefly explain the following terms taken from particle swarm optimisation, and their purpose in the algorithm:

- (i) *Velocity*
- (ii) *gbest*
- (iii) *pbest*

(6 marks)

- (f) Describe the differences in architecture between a *perceptron*, a *multilayer perceptron* and a *self organising map* and the problem types that each neural network can be used for.

(8 marks)

(Total 40 marks)

Please Turn Over

2. (a) Nature-Inspired Algorithms must strike a balance between *exploration* and *exploitation*. Explain what this means in terms of a search and optimisation problem, illustrating your answer with an example of an algorithm that is largely exploitative and one which is largely exploratory.

(15 marks)

Each nature-inspired algorithm has a different method of varying the extent to which it *exploits* or *explores*. For each algorithm below, describe the parameters that are modified to achieve different exploitation and exploration capability and explain how varying these alters the behaviour of the algorithm.

- i. Genetic Algorithms
- ii. Particle Swarm Optimisation
- iii. Ant Colony Optimisation

(15 marks)

(Total 30 marks)

Please Turn Over

3. Ant colony optimisation is a successful algorithm that can be used for a large number of optimisation problems, including those that do not have an explicit topology. One such problem is that of water distribution optimisation where each pipe from the set of pipes p needs to be assigned a single diameter chosen from the finite set d . The network will be simulated and a pressure value will be given for each node, this pressure must be equal to or greater than 15.0 for each node in the network. The cost of each pipe is equal to $pipeDiameter * pipeLength$.

(a) Draw the *construction graph* needed to solve this problem using ant colony optimisation.

(8 marks)

(b) Explain how you would calculate the objective function to find a least cost network that meets or exceeds the pressure requirements.

(8 marks)

(c) Describe the parameter settings you would initially use to solve the problem using ant colony optimisation.

(8 marks)

(d) Describe the experiments you would undertake to determine whether your initial assumptions about the parameters are correct.

(6 marks)

(Total 30 marks)

Please Turn Over

4. Neural networks are useful pattern recognition tools and can also be used to study human cognition.

- (a) Describe the neural network approach (including representation), and briefly the experimental findings in the paper Sejnowski, T. J. and Rosenberg, C. R. (1987) "Parallel networks that learn to pronounce English text" *Complex Systems* **1**, 145-168.

(15 marks)

(b) Using both equations and text, explain how a multilayer perceptron:

- i. Calculates an output given a set of input data.

(7 marks)

- ii. Calculates the error on the output and updates the weights to generate a closer output in the next iteration.

(8 marks)

(Total 30 marks)

Please Turn Over

5. (a) Genetic programming shares some of its features with genetic algorithms, but there are some key differences between the two methods. Discuss the differences between the two in terms of:
- i. The representation of solutions
 - ii. The creation of random individuals
 - iii. The methods used to vary the solutions

(15 marks)

- (b) You have been given a set of results of a marketing survey which shows how many people enter a restaurant at a particular time along with other data relating to the prevailing weather and the promotions being offered. The managers think that the number of people entering the restaurant is dependent on the time of day, the outside temperature and whether or not they have their “2 for 1” offer running. They have asked you to find the relationship using genetic programming.

There are five steps to be taken before running genetic programming on a problem. Describe each step and for each step, show how you would set up the genetic programming algorithm to solve the above problem.

(15 marks)

(Total 30 marks)

End of Paper